

Tree Height Triangulation Shootout

Measure a tree without climbing it.

Win condition: Closest to instructor reference height.

THE SCIENCE YOU ARE USING

Angles give height. You can find a tree's height without touching the top. Sight the treetop through a paper clinometer to read an angle, measure your distance to the trunk, and the geometry of a right triangle gives the height. This is remote sensing with a protractor.

Accuracy from method. Small sighting errors grow over distance, so careful technique matters: level your eye, sight steadily, measure distance honestly. The team closest to the instructor's reference height wins.

HOW TO BUILD AND RUN IT

01 Build the clinometer. Fold a paper clinometer with a weighted string plumb line to read angles.

2 Use the marked distance. Go to the spot the instructor pre-marked with the long tape and record that fixed standoff distance from the trunk.

01 Sight the treetop. Look along the clinometer at the very top and read the angle steadily.

4 Compute the height. $\text{tree height} = \text{eye height} + (\text{distance} \times \tan(\text{angle}))$. Measure your eye height with the 12 in ruler and use the marked standoff distance for the distance.

01 Check against reference. Compare to the instructor reference. The closest estimate wins.

Safety: Outdoor boundaries. Allergy: Pollen sensitivity.

MATERIALS

Paper clinometers	4 (plus 2 spares)
String and washers	shared (1 roll string, 1 pack washers)
12 in rulers	4 (plus 2 spares; for eye height)
Long tape measure	1 (to pre-mark the standoff distance)
Clipboards	4 (plus 1 spare)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

Standoff distance from the trunk (marked by instructor): _____

Clinometer angle (average of several readings): _____

Eye height, and computed tree height = eye height + (distance x tan angle): _____

DEFEND YOUR ESTIMATE

What was your biggest source of error, and how close were you to the reference? _____

If you measured again, what is the ONE thing you would change to be more accurate? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	100